Toy DMP Model in CT

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This note follows a lot from the note by Toshihiko Mukoyama.

1 Unemployment Dynamics

Let the period length be Δ and unemp. rate at time t be $u(t)$:

$$u(t+\Delta) = \sigma\Delta(1-u(t)) + (1-\lambda_w\Delta)\,u(t)$$

where $\sigma\Delta > 0$ is the prob. of losing the job, $\lambda_w\Delta > 0$ be the prob. of finding a job over Δ.

We can rewrite it as:

$$\begin{aligned}
u(t+\Delta) &= \sigma\Delta - \sigma\Delta u(t) + u(t) - \lambda_w\Delta u(t) \\
u(t+\Delta) - u(t) &= \sigma\Delta - \sigma\Delta u(t) - \lambda_w\Delta u(t) \\
\frac{u(t+\Delta)-u(t)}{\Delta} &= \sigma - \sigma u(t) - \lambda_w u(t) \\
\frac{u(t+\Delta)-u(t)}{\Delta} &= \sigma(1-u(t)) - \lambda_w u(t)
\end{aligned}$$

so taking $\Delta \to 0$:

$$u'(t) = \sigma(1-u(t)) - \lambda_w u(t)$$

We then try to endogenize job-finding rate λ_w. Let agg. vacancy be v and the matching function takes the form of

$$M(u,v)\,\Delta$$

Some assumptions:

1. matching is random
   (a) all vacancies have an equal chance to match w/ an employed worker
   (b) all unemployed workers have an equal chance to match w/ a vacancy
2. M is increasing in both terms
3. M is CRS

Under these assumptions, the prob. of a vacancy matching w/ an unemp. worker during Δ is

$$\frac{M(u,v)\Delta}{v} = M\left(\frac{1}{\theta},1\right)\Delta = \lambda_f(\theta)\Delta$$

where $\theta \equiv \frac{v}{u}$ is the tightness. The prob. of an unemp. worker matching w/ a vacancy during Δ is:

$$\frac{M(u,v)\Delta}{u} = M(1,\theta)\Delta = \lambda_w(\theta)\Delta$$

and $\lambda_w(\theta) = M(1,\theta) = \theta M\left(\frac{1}{\theta},1\right) = \theta\lambda_f(\theta)$. So unemp. rate dynamics now become:

$$u'(t) = \sigma(1-u(t)) - \lambda_w(\theta(t))\,u(t)$$

1.1 Beveridge curve condition

Consider ss. whre $v(t) \equiv v, \forall t$ and $u'(t) = 0$:

$$\begin{aligned} 0 &= \sigma(1-u) - \lambda_w\left(\frac{v}{u}\right)u \\ \lambda_w\left(\frac{v}{u}\right)u &= \sigma(1-u) \\ u\left(\sigma + \lambda_w\left(\frac{v}{u}\right)\right) &= \sigma \\ u &= \frac{\sigma}{\sigma + \lambda_w\left(\frac{v}{u}\right)} \end{aligned}$$

This $u = \frac{\sigma}{\sigma+\lambda_w\left(\frac{v}{u}\right)}$ is referred to as the Beveridge curve condition.

2 DMP Model

2.1 Workers and Firms

The DMP builds on the unemp. dynamics and adds the endo. determination of agg. vacancy $v(t)$. Firms create vacancies by trading off the current vacancy posting cost and future profit from the firm-worker match.

We assume that each vacancy hires one worker. Let $w(t)$ be the eq. wage. The PV of being employed, in discrete-time setting w/ period length Δ is

$$W(t) = w(t)\Delta + \frac{1}{1+r\Delta}\left((1-\sigma\Delta)W(t+\Delta) + \sigma\Delta U(t+\Delta)\right) \tag{1}$$

where $U(t)$ is the value of being unemployed.

$$U(t) = b\Delta + \frac{1}{1+r\Delta}\left(\lambda_w(\theta(t))\Delta W(t+\Delta) + (1-\lambda_w(\theta(t))\Delta)U(t+\Delta)\right)$$

Multiplying $1+r\Delta$ on both sides of (1) we get:

$$\begin{aligned}
(1+r\Delta)W(t) &= (1+r\Delta)w(t)\Delta + \left((1-\sigma\Delta)W(t+\Delta) + \sigma\Delta U(t+\Delta)\right)\\
r\Delta W(t) &= (1+r\Delta)w(t)\Delta + \left((1-\sigma\Delta)W(t+\Delta) + \sigma\Delta U(t+\Delta)\right) - W(t)\\
rW(t) &= (1+r\Delta)w(t) + \sigma U(t+\Delta) + \left(\frac{1}{\Delta} - \sigma\right)W(t+\Delta) - \frac{1}{\Delta}W(t)\\
rW(t) &= (1+r\Delta)w(t) + \sigma\left(U(t+\Delta) - W(t+\Delta)\right) + \frac{W(t+\Delta) - W(t)}{\Delta}
\end{aligned}$$

so taking $\Delta \to 0$:

$$rW(t) = w(t) + \sigma\left(U(t) - W(t)\right) + W'(t)$$

Similarly for $U(t)$:

$$\begin{aligned}
(1+r\Delta)U(t) &= (1+r\Delta)b\Delta + \left(\lambda_w(\theta(t))\Delta W(t+\Delta) + (1-\lambda_w(\theta(t))\Delta)U(t+\Delta)\right)\\
r\Delta U(t) &= (1+r\Delta)b\Delta + \left(\lambda_w(\theta(t))\Delta W(t+\Delta) + (1-\lambda_w(\theta(t))\Delta)U(t+\Delta)\right) - U(t)\\
rU(t) &= (1+r\Delta)b + \left(\lambda_w(\theta(t))W(t+\Delta) - \lambda_w(\theta(t))U(t+\Delta)\right) + \frac{U(t+\Delta) - U(t)}{\Delta}
\end{aligned}$$

so taking $\Delta \to 0$:

$$rU(t) = b + \lambda_w(\theta(t))(W(t) - U(t)) + U'(t)$$

Because of the one-firm to one-worker assumption, the firm is either matched or vacant, the value of a matched firm is:

$$J(t) = p(t)\Delta - w(t)\Delta + \frac{1}{1+r\Delta}(\sigma\Delta V(t+\Delta) + (1-\sigma\Delta)J(t+\Delta))$$

where $p(t)\Delta$is the product produced by the match and $V(t)$ is the value of vacancy, computed by:

$$V(t) = -\kappa\Delta + \frac{1}{1+r\Delta}\left(\lambda_f(\theta(t))\Delta J(t+\Delta) + \left(1-\lambda_f(\theta(t))\Delta\right)V(t+\Delta)\right)$$

where $\kappa\Delta$ is the cost of posting a vacancy. It's then easy to see that

$$rJ(t) = p(t) - w(t) + \sigma(V(t) - J(t)) + J'(t)$$

and

$$rV(t) = -\kappa + \lambda_f(\theta(t))(J(t) - V(t)) + V'(t) \tag{2}$$

We assume that anyone can start a firm by posting a vacancy

1. if $V(t) > 0$, new entrants would keep posting vacancies until $V(t)$ becomes 0. Additional vacancies would drop the value of $V(t)$because $\lambda_f(\theta(t))$ is a decreasing function of $\theta(t) = \frac{v(t)}{u(t)}$. Therefore $V(t) = 0$ has to always hold, as long as $v(t) > 0$. [1]

2. We call the condition

$$V(t) = 0$$

   the *free-entry condition*

One can formalize 1 as follows:

Let $u(t)$ be the state var. at time t, which evolves as

$$u' = \sigma(1-u) - \lambda_w u$$

[1] It's possible that $V(t) < 0$ when $v(t) = 0$.

and $v(t)$be the control var. and $v \geq 0$.Given (2) and taking $u(t), J(t), V'(t)$as given, we have

$$V(t) = \frac{\lambda_f(\theta(t)) J(t) - \kappa + V'(t)}{r + \lambda_f(\theta(t))}$$

so

$$\frac{\partial V}{\partial \lambda_f} = \frac{J(t)\left(r + \lambda_f(\theta(t))\right) - \left(\lambda_f(\theta(t)) J(t) - \kappa + V'(t)\right)}{\left(r + \lambda_f(\theta(t))\right)^2}$$

at ss., $V'(t) = 0$, so we have

$$\begin{aligned}\frac{\partial V}{\partial \lambda_f} &= \frac{rJ(t) + \kappa - V'(t)}{\left(r + \lambda_f(\theta(t))\right)^2} \\ &= \frac{rJ(t) + \kappa}{\left(r + \lambda_f(\theta(t))\right)^2} > 0\end{aligned}$$

Given the matching function, it's easy to see that $\lambda_f(\theta) = M\left(\frac{1}{\theta}, 1\right)$ is decreasing in θ:

$$\frac{\partial \lambda_f}{\partial v} = \lambda_f'(\theta)\frac{1}{u} < 0$$

so

$$\frac{\partial V}{\partial v} = \frac{\partial V}{\partial \lambda_f}\frac{\partial \lambda_f}{\partial v} < 0$$

Therefore, if $V(t) > 0$, then it follows that vacancy goes up and in turn pushing $V(t) \to 0$. If $V(t) < 0$, then nobody is willing to post a vacancy and in turn $v(t) = 0$. Finally if $V(t) = 0$, then entrants are indifferent and therefore $\forall v(t) \geq 0, s.t. V(v(t)) = 0$ will work.

2.2 Wage Determination

In this framework, for a worker to meet w/ another firm, the worker has to endure the cost of unemployment. Similarly, for the firm, losing a worker would imply waiting for another worker by incurring the vacancy cost. Thus, the matched worker and firm are in a bilateral monopoly situation.

Here we assume that the surplus is split by the level of wage that solves the following maximization problem:

$$\max_w \left(\tilde{W}(w,t) - U(t)\right)^\gamma \left(\tilde{J}(w,t) - V(t)\right)^{1-\gamma} \tag{3}$$

where $\gamma \in (0,1)$. The functions $\tilde{W}(w,t)$ and $\tilde{J}(w,t)$are the values of employed workers and a matched firm, here we make it explicit that only w in this period can move. The solution to this problem is often called *Generalized Nash Bargaining Rule.*

The functions $\tilde{W}(w,t)$ and $\tilde{J}(w,t)$ can be written as:

$$\tilde{W}(w,t) = w\Delta + \frac{1}{1+r\Delta}\left(\left(1-\sigma\Delta\right)W(t+\Delta) + \sigma\Delta U(t+\Delta)\right)$$

and

$$\tilde{J}(w,t) = p(t)\Delta - w\Delta + \frac{1}{1+r\Delta}\left(\sigma\Delta V(t+\Delta) + \left(1-\sigma\Delta\right)J(t+\Delta)\right).$$

By taking FOC for (3), we have:

$$\begin{aligned}
&\max_{w}\gamma\ln\left(\tilde{W}(w,t) - U(t)\right) + \left(1-\gamma\right)\ln\left(\tilde{J}(w,t) - V(t)\right) \\
&0 = \frac{\gamma}{\tilde{W}(w,t)-U(t)}\frac{\partial\tilde{W}(w,t)}{\partial w} + \frac{1-\gamma}{\tilde{J}(w,t)-V(t)}\frac{\partial\tilde{J}(w,t)}{\partial w}
\end{aligned}$$

Given $\frac{\partial\tilde{W}(w,t)}{\partial w} = \Delta$, $\frac{\partial\tilde{J}(w,t)}{\partial w} = -\Delta$, we have

$$\begin{aligned}
0 &= \frac{\gamma}{\tilde{W}(w,t)-U(t)}\Delta - \frac{1-\gamma}{\tilde{J}(w,t)-V(t)}\Delta \\
0 &= \frac{\gamma}{\tilde{W}(w,t)-U(t)} - \frac{1-\gamma}{\tilde{J}(w,t)-V(t)} \\
\frac{1-\gamma}{\tilde{J}(w,t)-V(t)} &= \frac{\gamma}{\tilde{W}(w,t)-U(t)} \\
\left(1-\gamma\right)\left(\tilde{W}(w,t)-U(t)\right) &= \gamma\left(\tilde{J}(w,t)-V(t)\right) \qquad (4)
\end{aligned}$$

so the w^*that satisfies (4) is the eq. wage.

2.3 Steady State Equilibrium

First, consider the ss. eq. where $u(t)$ and $v(t)$ are constant. With the conditions above we have

$$\begin{aligned} rW &= w - \sigma(W-U) \\ rU &= b + \lambda_w(\theta)(W-U) \\ rJ &= p - w - \sigma(J-V) \\ rV &= -\kappa + \lambda_f(\theta)(J-V) \\ V &= 0 \\ \left(1-\gamma\right)(W-U) &= \gamma(J-V) \end{aligned}$$

in total 6 equations w/ 6 unknowns W, U, J, V, θ, w. We can then try to reduce this system to 1 to 1. It can be shown that we have

$$(1-\gamma)(p-b) - \frac{r+\sigma+\gamma\lambda_w(\bar{\theta})}{\lambda_f(\bar{\theta})}\kappa = 0.$$

This condition is then used to solve for θ and is often referred to as the *job creation condition.*

For simple math, we have

$$\begin{aligned} rV &= -\kappa + \lambda_f(\theta)(J-V) \\ 0 &= -\kappa + \lambda_f(\theta)(J-0) \\ J &= \frac{\kappa}{\lambda_f(\theta)} \end{aligned}$$

Given

$$\begin{aligned} rJ &= p - w - \sigma(J-V) \\ rJ &= p - w - \sigma J \\ J &= \frac{p-w}{r+\sigma} \end{aligned}$$

we have the free-entry condition on firm side:

$$\frac{\kappa}{\lambda_f(\theta)} = \frac{p-w}{r+\sigma}$$

Given

$$\begin{aligned}
rW &= w - \sigma(W-U) \\
rU &= b + \lambda_w(\theta)(W-U) \\
r(W-U) &= w - b - (\sigma + \lambda_w(\theta))(W-U) \\
(r+\sigma+\lambda_w(\theta))(W-U) &= w-b \\
W-U &= \frac{w-b}{r+\sigma+\lambda_w(\theta)}
\end{aligned}$$

so given $\left(1-\gamma\right)(W-U) = \gamma(J-V)$, we have

$$\begin{aligned}
\left(1-\gamma\right)\frac{w-b}{r+\sigma+\lambda_w(\theta)} &= \gamma\frac{\kappa}{\lambda_f(\theta)} \\
w-b &= \frac{\gamma}{1-\gamma}\frac{\kappa}{\lambda_f(\theta)}(r+\sigma+\lambda_w(\theta)) \\
p-w &= p-\left[\frac{\gamma}{1-\gamma}\frac{\kappa}{\lambda_f(\theta)}(r+\sigma+\lambda_w(\theta))+b\right] \\
(r+\sigma)\frac{\kappa}{\lambda_f(\theta)} &= p-\left[\frac{\gamma}{1-\gamma}\frac{\kappa}{\lambda_f(\theta)}(r+\sigma+\lambda_w(\theta))+b\right] \\
(r+\sigma)\kappa &= \lambda_f(\theta)\left(p-b\right) - \frac{\gamma\kappa}{1-\gamma}(r+\sigma+\lambda_w(\theta)) \\
(r+\sigma)\kappa + \frac{\gamma\kappa}{1-\gamma}(r+\sigma+\lambda_w(\theta)) &= \lambda_f(\theta)\left(p-b\right) \\
\left(1-\gamma\right)(r+\sigma)\kappa + \gamma\kappa(r+\sigma+\lambda_w(\theta)) &= \left(1-\gamma\right)\left(p-b\right)\lambda_f(\theta) \\
(r+\sigma)\kappa + \gamma\kappa\lambda_w(\theta) &= \left(1-\gamma\right)\left(p-b\right)\lambda_f(\theta) \\
\frac{r+\sigma+\gamma\lambda_w(\theta)}{\lambda_f(\theta)}\kappa &= \left(1-\gamma\right)\left(p-b\right)
\end{aligned}$$

which is exactly the job creation condition. To explain this:

$$\underbrace{\frac{r+\sigma+\gamma\lambda_w(\theta)}{\lambda_f(\theta)}}_{\text{annuitized job creation cost}}\kappa = \underbrace{(1-\gamma)(p-b)}_{\text{firm's share of match surplus}}$$

2.4 Transition Dynamics

Now let's consider the transition dynamics by not imposing ss. conditions. Collect:

$$\begin{aligned}
rW(t) &= w(t)+\sigma(U(t)-W(t))+W'(t)\\
rU(t) &= b+\lambda_w(\theta(t))(W(t)-U(t))+U'(t)\\
rJ(t) &= p(t)-w(t)+\sigma(V(t)-J(t))+J'(t)\\
rV(t) &= -\kappa+\lambda_f(\theta(t))(J(t)-V(t))+V'(t)\\
(1-\gamma)\left(\tilde{W}(w,t)-U(t)\right) &= \gamma\left(\tilde{J}(w,t)-V(t)\right)\\
V(t) &= 0
\end{aligned}$$

so

$$\begin{aligned}
0 &= -\kappa+\lambda_f(\theta(t))(J(t)-0)\\
J(t) &= \frac{\kappa}{\lambda_f(\theta(t))}
\end{aligned}$$

and

$$\begin{aligned}
J'(t) &= rJ(t)-\left[p(t)-w(t)+\sigma(V(t)-J(t))\right]\\
&= (r+\sigma)J(t)-\left[p(t)-w(t)\right]\\
&= (r+\sigma)\frac{\kappa}{\lambda_f(\theta(t))}-p(t)+w(t)
\end{aligned}$$

also given $(1-\gamma)(W(t)-U(t))=\gamma(J(t)-V(t))$ holds, we have that

$$\begin{aligned}
(1-\gamma)\left(W'(t)-U'(t)\right) &= \gamma\left(J'(t)-V'(t)\right)\\
(1-\gamma)\left(W'(t)-U'(t)\right) &= \gamma J'(t)
\end{aligned}$$

and

$$
\begin{aligned}
W'(t) - U'(t) &= rW(t) - [w(t) + \sigma(U(t) - W(t))] - rU(t) + b + \lambda_w(\theta(t))(W(t) - U(t)) \\
&= (r + \lambda_w(\theta(t)) + \sigma)(W(t) - U(t)) - w(t) + b
\end{aligned}
$$

so

$$
\begin{aligned}
\frac{\gamma J'(t)}{1-\gamma} &= (r + \lambda_w(\theta(t)) + \sigma)\frac{\gamma}{1-\gamma}\frac{\kappa}{\lambda_f(\theta(t))} - w(t) + b \\
\frac{\gamma J'(t)}{1-\gamma} + J'(t) &= \left[(r + \lambda_w(\theta(t)) + \sigma)\frac{\gamma}{1-\gamma}\frac{\kappa}{\lambda_f(\theta(t))} - w(t) + b\right] + \left[(r+\sigma)\frac{\kappa}{\lambda_f(\theta(t))} - p(t) + w(t)\right] \\
\frac{J'(t)}{1-\gamma} &= \frac{r + \sigma + \gamma\lambda_w(\theta(t))}{1-\gamma}\frac{\kappa}{\lambda_f(\theta(t))} + b - p(t) \\
J'(t) &= \frac{r + \sigma + \gamma\lambda_w(\theta(t))}{\lambda_f(\theta(t))}\kappa - (1-\gamma)[p(t) - b]
\end{aligned}
$$

I've been checking this and it seems that Toshihiko's note is having the wrong sign here. Anyway let's move on. Given $J(t) = \frac{\kappa}{\lambda_f(\theta(t))}$ we have

$$
\begin{aligned}
J(t) &= \frac{\kappa}{\lambda_f(\theta(t))} \\
J'(t) &= \frac{-\lambda'_f(\theta(t))\theta'(t)}{[\lambda_f(\theta(t))]^2} \\
&= \frac{\kappa}{\lambda_f(\theta(t))}\underbrace{\left(-\frac{\lambda'_f(\theta(t))\theta(t)}{\lambda_f(\theta(t))}\right)}_{\equiv\eta(\theta(t))}\frac{\theta'(t)}{\theta(t)} \\
&= J(t)\eta(\theta(t))\frac{\theta'(t)}{\theta(t)}
\end{aligned}
$$

where $\eta(\theta(t)) = -\frac{\lambda'_f(\theta(t))\theta(t)}{\lambda_f(\theta(t))} > 0$ is the elasticity of matching function. So if $\theta(t) > \theta$ where θ is the ss. value of the tightness, we have $J'(t) > 0$ so that $\theta'(t) > 0$, similarly if $\theta(t) < \theta$, we have $\theta'(t) < 0$. So that the only case where $\theta(t)$ does converge is that $\theta(t) \equiv \theta$, then it's easy to see that $J'(t) = 0$ and $J(t)$ is fixed too.

Therefore, we can say that job-creation condition holds even not in ss. So given

$$\theta = \frac{v(t)}{u(t)}$$
$$\frac{r+\sigma+\gamma\lambda_w(\theta)}{\lambda_f(\theta)}\kappa = \left(1-\gamma\right)\left(p-b\right)$$
$$u'(t) = \sigma(1-u(t)) - \lambda_w(\theta(t))\, u(t)$$

one can pin down the dynamics of $(v(t), u(t), \theta)$. Now let's consider the case where $u(t) > u$, where u is given by the Beveridge curve condition:

$$u = \frac{\sigma}{\sigma + \lambda_w\left(\frac{v}{u}\right)}$$

then we have $u'(t) < 0$, similarly we have $u'(t) > 0$ when $u(t) < u$, so it's easy to establish that $u(t) \to u$ monotonically. Given $v = \theta u$, the dynamic of $v'(t)$ is similar.